दक्षिण बिहार केन्द्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination May 2025

Name of Department: Department of Computer Science **Programme: M. Sc. (AI)**

Session: 2024-2026 **Semester: 2nd**

Course Title: Wireless Sensor Networks **Course Code: CAI82DE01304**

Course Credit: 4 **Max. Points: 70** **Duration: 3:00 Hrs.**

Enrolment No. (to be filled by Student) ....

Instruction:

1. Attempt questions in all the Sections as directed.

Section - A

Answer all questions from below, each carrying 1 point. *(10 X 1 = 10)*

1. What are the differences between analog sensors and digital sensors?
2. What features should a sensor node have to build practical applications? List them.
3. What is the use of RSS in WSN?
4. Define coverage in the context of WSN.
5. WSNs are data-centric networks. What does it mean?
6. List the advantages of using clustering in WSN.
7. What are the key advantages of using SPIN for data dissemination in WSN?
8. What is the role of data aggregation in WSN?
9. List the key operations that consume energy in a sensor node's radio unit.
10. List any three potential sources of energy waste in a WSN MAC protocol.

Section - B

Answer any 5 questions below, each carrying 6 points. *(5 X 6 = 30)*

1. Discuss how WSN is different from Wired Networks.
2. Elaborate on the techniques that can be used in a node discovery protocol.
3. Discuss a typical architecture of a Wireless Sensor Network and its elements.
4. What is gossiping and how does it differ from flooding? Describe the limitations of flooding. [2+2+2]

5. What are the types of coverage? What challenges you may encounter in obtaining the desired level of coverage during a WSN deployment? [2+ 4]

6. What is virtual carrier sensing? Explain.

Section - C

Answer any 2 questions below, each carrying 15 Points. ***(2 X 15 = 30)***

1. What do you understand by query-based routing and cluster-based routing protocols? What are the different challenges in designing routing protocols in WSN? Explain. [7.5+ 7.5]

2. Write a short note on any two of the following, each carrying 7.5 points:

   a. Classification of Sensor Node Deployment Techniques

   b. Power Consumption Model in WSN

   c. Classification of various localization techniques

3. Answer the following, each carrying 7.5 points:

   a. Explain the security challenges associated with Wireless Sensor Networks. Provide application-based examples also while discussing the challenges.

   b. Explain the unique challenges associated with MAC protocols in WSN.